

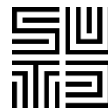
Manufacturing and Service Operations

MSO 40.012

R Desmicht – Adjunct Professor SUTD

Agenda Course 1

- Course Introduction:
 - What is operation management
 - Why is it important
 - What is the scope
- Long-term Operation Management strategic focus
 - Product characteristics
 - Segmentation
 - Line concept
 - Capacity Management



About myself



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Whatsapp: 85050216

Linkedin

China European Business School MBA

Professional Coach

Asia

Solid State Physics

Leadership

Nobel Prize

Paris

Learn and explore

Reading

Infineon

Travel the World

普通话

Grandes Ecoles

ams OSRAM

Muay Thai

Amkor Technology

Singapore

Teamwork

Supply Chain

People development

Nano-technologies

Diversity

Family

Thalès

James Bond

Shanghai

About our TA: CS03



About our TA: CS02



About our TA: CS01:





Teaching method and expectations

Process Focus over math

Class Questions/Quizz

Group Work

Be inquisitive

Try it yourself

Class attendance and
Active participation

Lectures

Real life experience

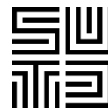
Self-reflection

Everyone contributes to other's learning

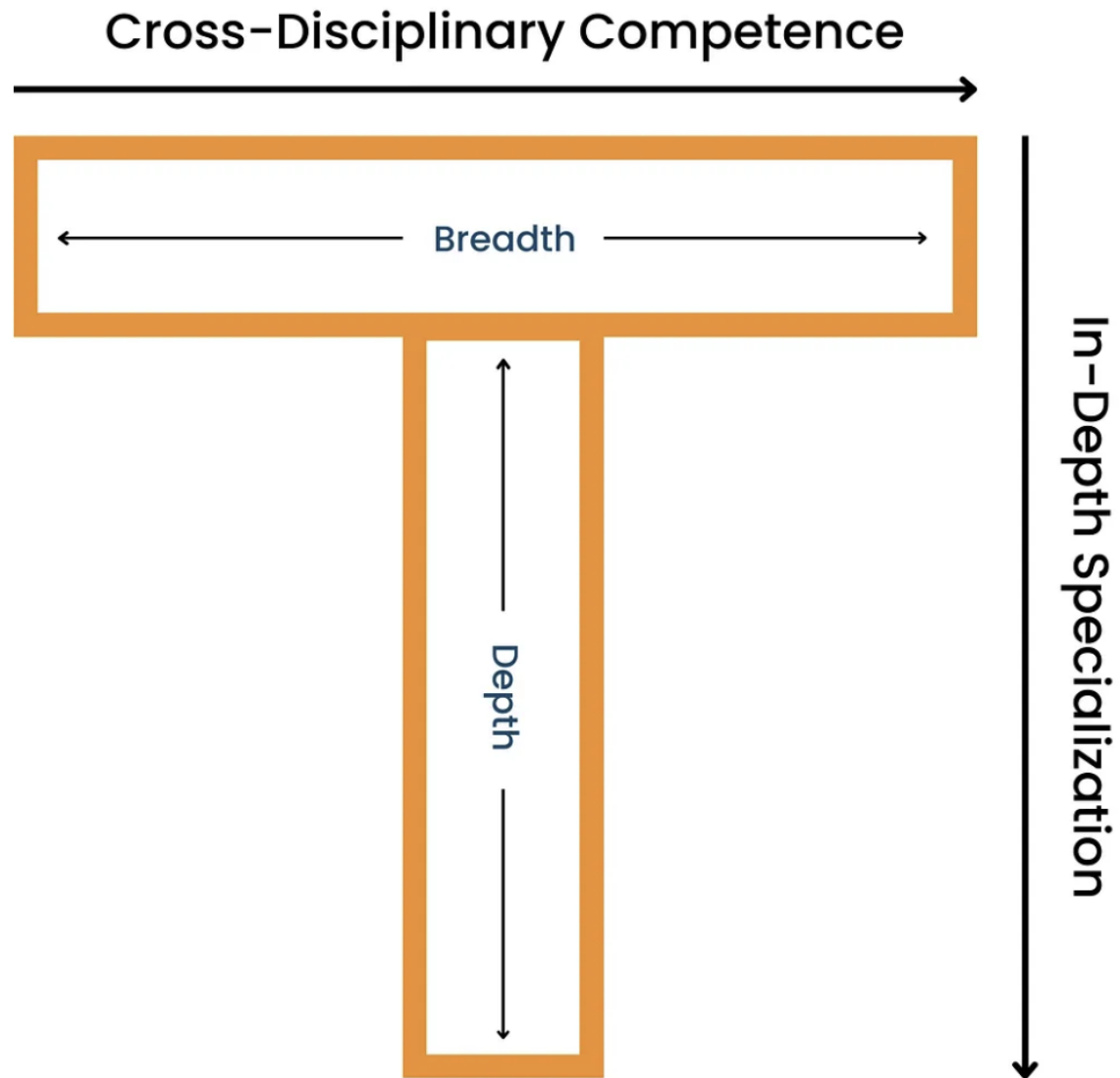
Exercises

All questions are good questions

3 Class to class learning takeaways



T Shape Learning



MSO Part 1

The field of Manufacturing Operations is BROAD



Course outline Part 1



Week(s)	Title	Instructor
Week 1	Introduction to MSO Operations Strategy	Roxane Desmicht
Week 2	Forecasting Aggregate Planning	Roxane Desmicht
Week 3	Master Production Schedule Material Requirement Planning	Roxane Desmicht
Week 4	Scheduling Project Management	Roxane Desmicht
Week 5	Inventory Management known demand Inventory Management unknown demand	Roxane Desmicht
Week 6	Quality Management Project Presentation Mid Term exam	Roxane Desmicht

MSO Part 1 and Part 2 are very different in content and focus.

Bibliography part 1

- ***Production and Operations Analytics, Eighth Edition***, by Steven Nahmias, Tava Lennon Olsen ISBN 10: 1478639261 / ISBN 13: 9781478639268; Published by Waveland Pr Inc, 2020 (or any previous ed.)
- ***Principles of Operations Management, Sustainability and Supply Chain Management, Twelfth Edition***, by Jay Heizer, Barry Rendered, Chuck Munson; Published by Pearson Education
- ***(Optional) Factory Physics*** by W. J. Hopp and M. L. Spearman.

LEGO workshop: mandatory

- Details: May 20:
 - CS03: 2pm to 3pm
 - CS02: 3:30pm to 4:30pm
 - CS01: 5pm to 6pm
- Report out in week 6: details to be found in eDimensions
- Pre-reading



Adobe Acrobat
Document

Seminar

- AI in Forecasting: May 26 6:30pm to 7:30pm LT4
- Scheduling Production: June 16 6:30pm to 7:30pm LT4
- No recitation in week 1
- Make-up classes:
 - CS01: 29 May 2026 (Fri), 9am - 11am, CC14
 - CS01: 2 Jun 2026 (Tue), 10.30am - 12.30pm, CC14
 - LS01: 4 Jun 2026 (Thu), 11.30am - 12.30pm, LT4

Project



- The project work will act as a recap for the whole course as it will cover most of the subject matters we studied.
- It will help you to bridge between academic knowledge and real life scenario.
- It will provide you with a hands-on experience on our various subject matters
- For some topics you might need to pre-study them ahead of the class as you might start the project earlier than the class schedule.
- You might book consultation time if guidance are needed.

Project

- Predefined teams of 4 students
- 2 teams per topic
- Report out in ppt format to be presented during the last class
- The project involves identifying a company to host you and share some information/data with you
- Presentation time: max 15min
- The presentation should cover:
 - Introduction (company and topic)
 - Problem formulation
 - Approach and link with the course
 - Numerical study/application if relevant
 - Discussion/benefits
 - Conclusion and key learnings
 - References as annex

Project Topics



Forecasting

Aggregate
Planning

Inventory
Control

MRP

Project
Management

Project Topics

- Forecasting:
 - Select a manufacturing or service company
 - Review with them their current forecasting method
 - Identify the factors they take into account while forecasting
 - Critique the method
 - If the company is willing to share data for one or two products apply the technique we studied and evaluate the associated error
 - Compile your recommendations on the technique the company should use and the benefits
- Aggregate Planning:
 - Select a manufacturing company
 - Map their supply chain (including tier 1 suppliers, manufacturing suppliers and distribution network)
 - Review with them their current aggregate planning approach; evaluate their maturity
 - Identify their method for aggregating products into groups; discuss the pros/cons
 - If they are willing to share data develop your own aggregate plan using one of the method we studied
 - Compile your recommendations on the aggregation schemes, factors they might consider, rolling horizon, ... Anticipate the benefits

Project Topics

- Inventory control:
 - Select a manufacturing or service company for which the demand is either known or unknown
 - Review with them their current inventory control policy
 - Understand the inventory decision for different products
 - Critique the method and suggest alternative
 - Try to evaluate the impact of your alternative
 - Compile your recommendations and the benefits
- MRP:
 - Select a manufacturing company with a robust ERP
 - Select 2 or 3 products, identify the BOM, critical suppliers and draw the product tree
 - Discuss the settings for their MRP runs for different types of raw materials
 - Understand what works well in the MRP runs, what would need improvement, how often planning parameters are reviewed, ...
 - Compile your recommendations. Anticipate the benefits

Project Topics

- Project Management:
 - Select a contractor involved in project work (building, consulting, ...)
 - Select one of their critical project
 - Explain how they worked out the timeline
 - Using the method we studied re-validate their timeline
 - Explain how they track project progress and manage risks
 - Provide an overview of their previous projects completion track record and your analysis of the results. Suggest recommendations.

Assessment

Assessment Part 1

Class Participation and quizzes (10%), Project (20%), Lego Game Report (10%), Midterm Exam (25%), Final Exam (25%), Homework (10%)

△ Class attendance $\neq$ Class participation

On time => 3 points

Late but before break => 2 points

Late after break => 1 point

Absent without excuse => 0 point

Bonus: 1 to 2 points based on quizzes and observed participation in class discussion

Plagiarism and Cheating

All parties engaging in plagiarism or cheating on examinations or assignments will receive a zero grade for the relevant work and serious disciplinary actions might ensue.

Use of AI

AI is a Tool, Not the Author

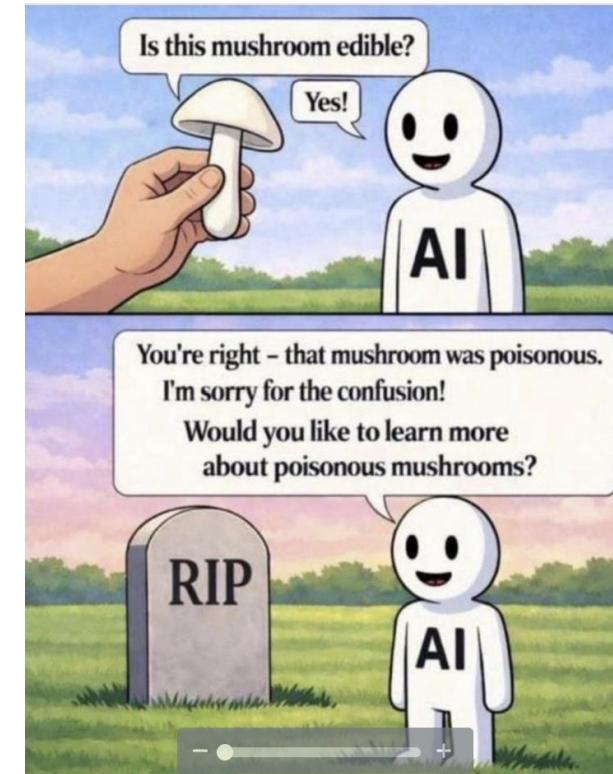
- Use AI for brainstorming or outlines, but final analysis must be yours
- Your unique perspective and critical thinking earn the grade

Master the Art of Citation – If You Didn't Write It, Cite It

- Declare which AI tools you used and how you used them
- If an idea/phrase came from a prompt, give credit just like any other source (APA style)
- Save your prompts and drafts as proof of your original process

Fact-Check Everything

- AI often “hallucinates” (makes up facts or fake citations)
- If the AI cites a book or article, confirm it exists before including it



Mid-term exam

Closed booked

1 A4 double-sided hand-written cheat sheet

5 to 6 **independent** problems related to each section of the course

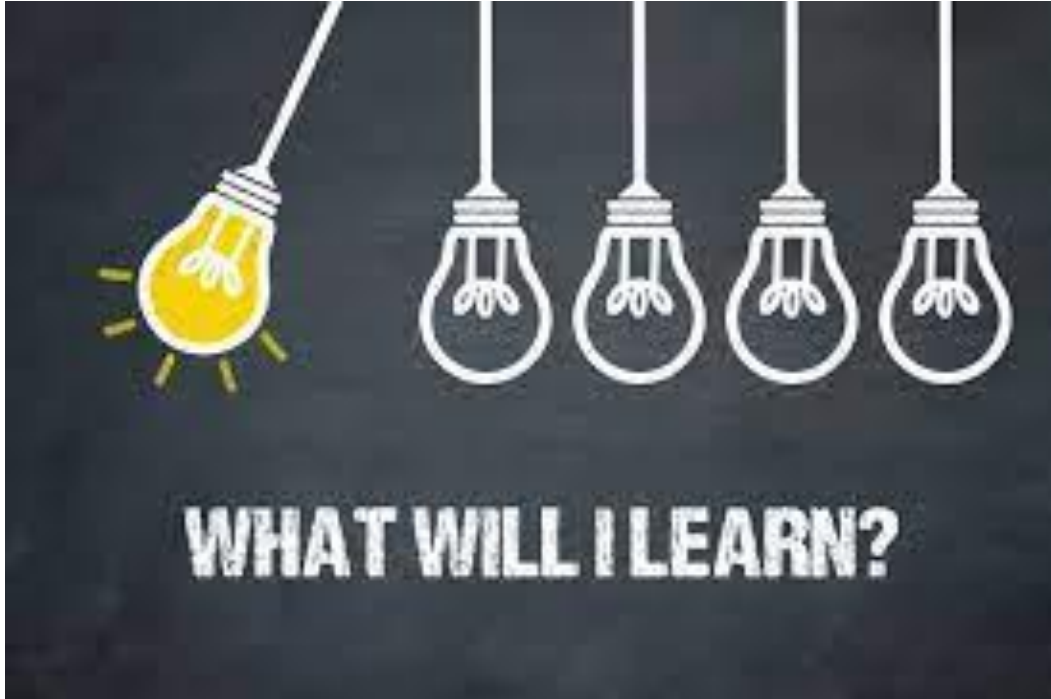
Each problem will be similar to the exercises worked out in class.

Last year exam for reference.



Adobe Acrobat
Document

Learning Objectives



Define MSO

Describe the role of MSO within the firm

Clarify key MSO responsibilities

Explore an example

Take stock of the current knowledge

What is operations management



Production is the creation of goods and services



Operations management (OM) is the set of activities that creates value in the form of goods and services by transforming inputs into outputs as efficiently as possible. Part 1 focuses on goods, part 2 on services.

More and more CEOs coming out of operations

- Apple => Tim Cook
- GM=> Mary Barra
- P&G=> David Taylor
- Walmart=> Doug McMillon
- Intel=> Brian Krzanich
- Nike=> Mark Parker
-



- Out of the World Economic Forum: 'performance results and shareholder value dépend heavily on understanding the interaction of money, markets and materials'.

What do you know about it?



Quizz Time

Organisation of Business

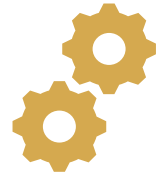
Marketing and sales

- Assessing customer wants and needs
- Promoting and Selling
- Communicating needs to operations



Operations/Production

- Goods oriented (manufacturing and assembly)
- Service oriented (health care, transportation and retailing)
- Value-added (the essence of the operations functions)



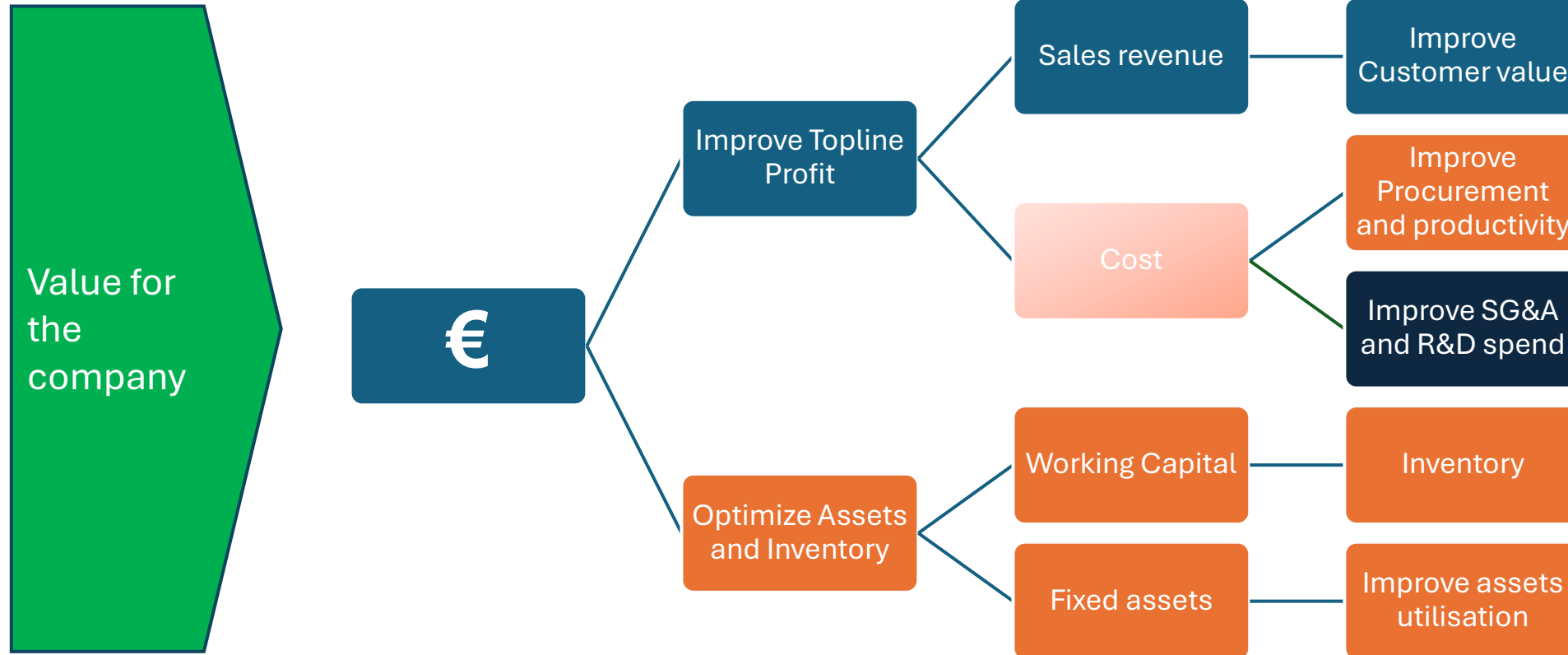
Finance

- Budgets (plan financial requirements)
- Economic analysis of investment proposals
- Provision of funds (the necessary funding of the operations)



Why is it important?

Link Operations & Supply Chain with Business Value



Why is it important?



OM is one of three major functions of any organization



OM is a costly part of an organization



We want (and need) to know how goods and services are produced



We want to understand what operations managers do



We want to study how people organize themselves for a productive enterprise



We want to study how an enterprise can improve

What is our scope?

1

■ Strategy

- Capacity choice
- Production Site
- Products or services
- Make or buy
- Location strategy
- Risk control

2

■ Tactical

- Inventory management
- Flow management
- Forecast
- Planning
- Quality management

3

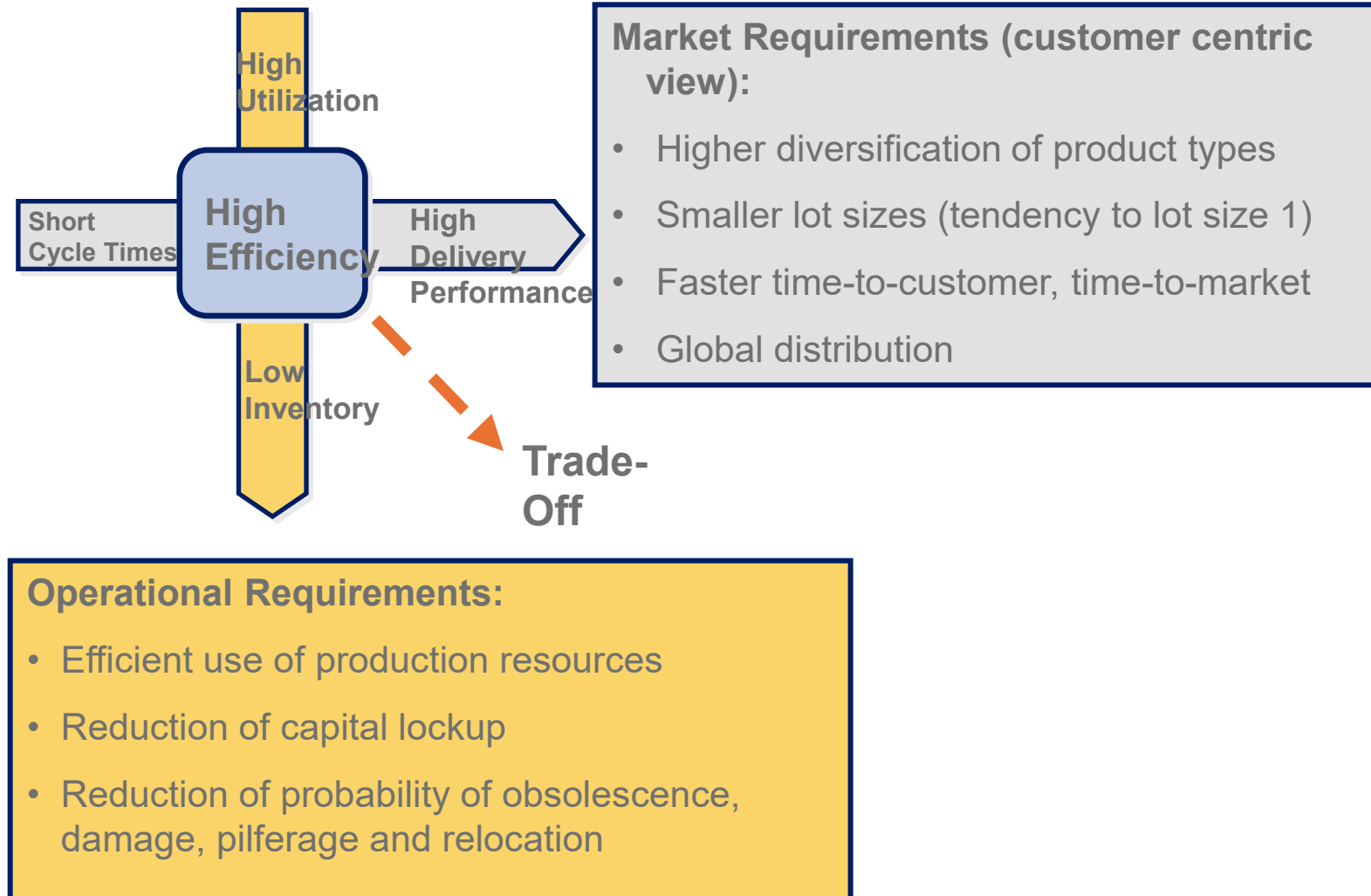
■ Execution and control

- Scheduling
- Buying
- Quality control
- Cost control
- Maintenance

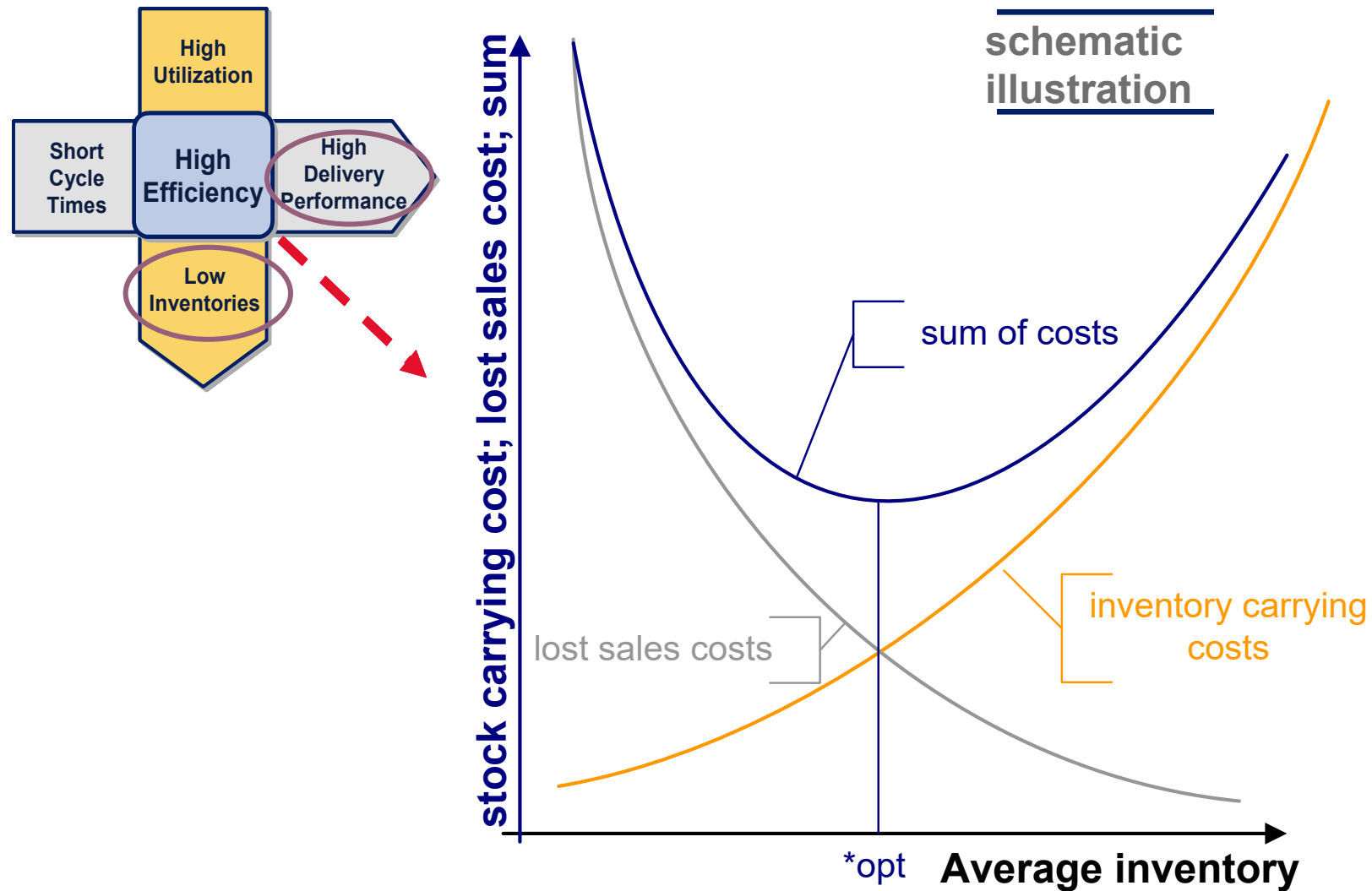
To make relevant decisions, we need to **segment** different operations models, aligned with corporate strategy.

Note: operations models (should) correspond to business models.

Our bread and butter: managing trade-offs



Trade-off example



Evolution through time

- Early 1900s : full vertical integration
- Ford River Rouge Plant :
 - Ford owned the railroad
 - Ford owned the electricity power
 - Ford owned the steel mill
 - Everything needed to manufacture the car was there
- **And it would produce at the early stage the model T in black**



Aerial view of the Rouge Complex in 1927

To the 21st century and the 4th industrial revolution: I4.0

3 pillars:

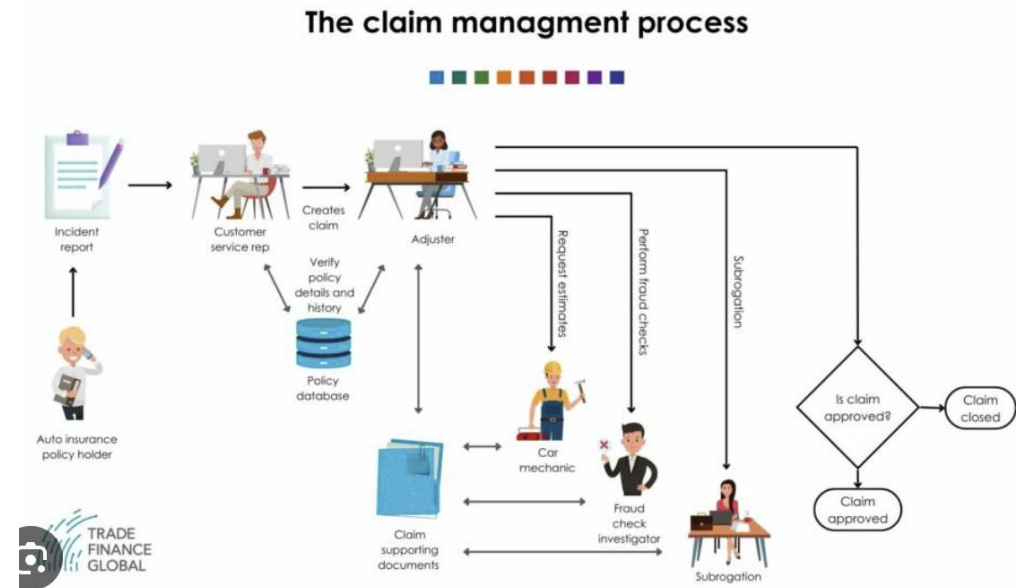
- Vertical integration : only within the 4 walls of the factory; breakthrough with automation
- Horizontal integration along the whole network from supplier's supplier to customer's customers with a continuous flow of communication and collaboration
- Digital integration where products optimize themselves based on previous stage results



Some examples



- <https://youtu.be/h4rdulAGbbQ?si=NgTs87hAbB4HI77a>
- https://youtu.be/TJ-PWomH1n0?si=pi3VzcaHbNb3jr_e



Key decisions: an example

Management of the conversion process which **transforms** inputs such as raw material and labor into outputs in the form of finished goods and services.



Where do we produce:

Close to the forest? In low cost country? In our own factory?

What do we produce:

Do we own forests or buy the wood?

How do we produce:

Do we have the same manufacturing line for the chair and table?

Do we keep inventory?

Which equipment do we need? How long does it take?

How much do we need to produce

What is the demand? How much capacity do we need? How many people?

How many machines?



Typical topics we have to tackle

Decision 1: Where and what

- Make or Buy/Outsourcing
- **Manufacturing footprint**
- Logistics footprint (storage and transport)

Decision 2: How

- **Production system/Line concept**
- Equipment type
- **Shared resources or dedicated**
- **Chase or Level**

Decision 3: How Much

- **Inventory Management**
- # equipment
- **MTS/MTO/...; aggregate planning, MPS**
- **Production scheduling**

Typical topics we have to tackle: exploration through an example

Kirsten's company



1. Start your analysis by creating a process diagram of the cookie manufacturing process.
2. Key questions:
3. How long will it take you to fill an urgent order of 12 cookies? Of 24 cookies of the same variety?
4. 2. How many orders can you fill in one night, assuming you are open for four hours each night?
5. How much of your precious time and your roommate's will it take to fill each order?
6. Since your baking trays can hold exactly a dozen cookies, you will produce and sell cookies by the dozen. Do you offer a discount to people who order two dozen of the same cookies, three dozen of the same cookies, or more? If so, how much?
7. How many electric mixers and baking trays will you need?
8. Are there any changes you can make to your production plans that will allow you to make better cookies or more cookies in less time or at a lower cost? For example, is there a bottleneck in your production process that you can cheaply develop? What is the effect of adding another oven?
9. What happens if you try to do the work alone?
10. When do you need to promise delivery? How can you quickly check your order sheet (list of current orders) and tell a caller when their order will be ready? How much of a safety margin for the schedule should be anticipated?
11. What other factors do you need to consider at this stage of planning your business?
12. Your product must be made to order as each order is potentially unique. If you decide to sell standard cookies instead, how would you change the production system? The order-taking process? Other policies?

Wrap-up



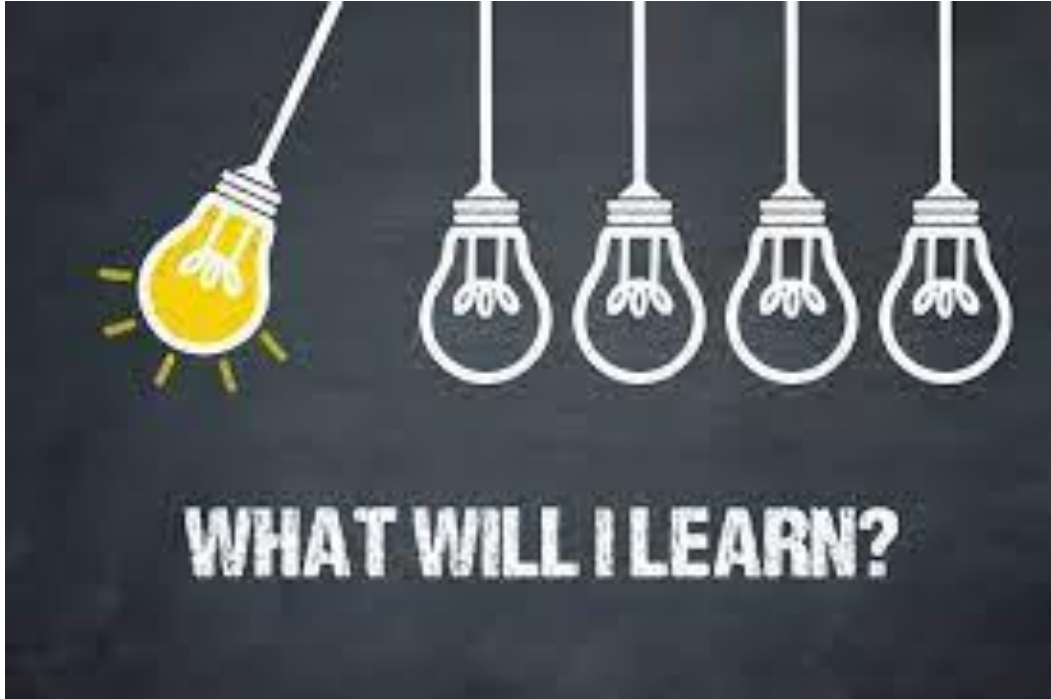
The case gets you to experience:

- The use of Gantt charts
- Bottleneck and their role
- Sizing the number of equipment you need
- Task optimization
- Idle resources
- Balance between customisation and equipment utilization
- Make to order versus Make To Stock

Agenda Course 1

- Course Introduction:
 - What is operation management
 - Why is it important
 - What is the scope
- Long-term strategic focus
 - Product characteristics
 - Segmentation
 - Line concept
 - Capacity Management

Learning Objectives



Identify how operations create a competitive advantage

Explain how goods and services are defined by operations

Describe process strategies

Explain the 4 major manufacturing segments

Define capacity and related indicators

Compute break even

Product Characteristics

Their USES and markets

- Consumables
- Components or a final product
- Equipment
- Unique or me too

Their economic impact

- Selling price
- Production cost
- Margin

Their features

- Shelf life
- Life span
- Traceability requirements
- Quantity and variability
- Number of variants

Product Characteristics

Perishable product

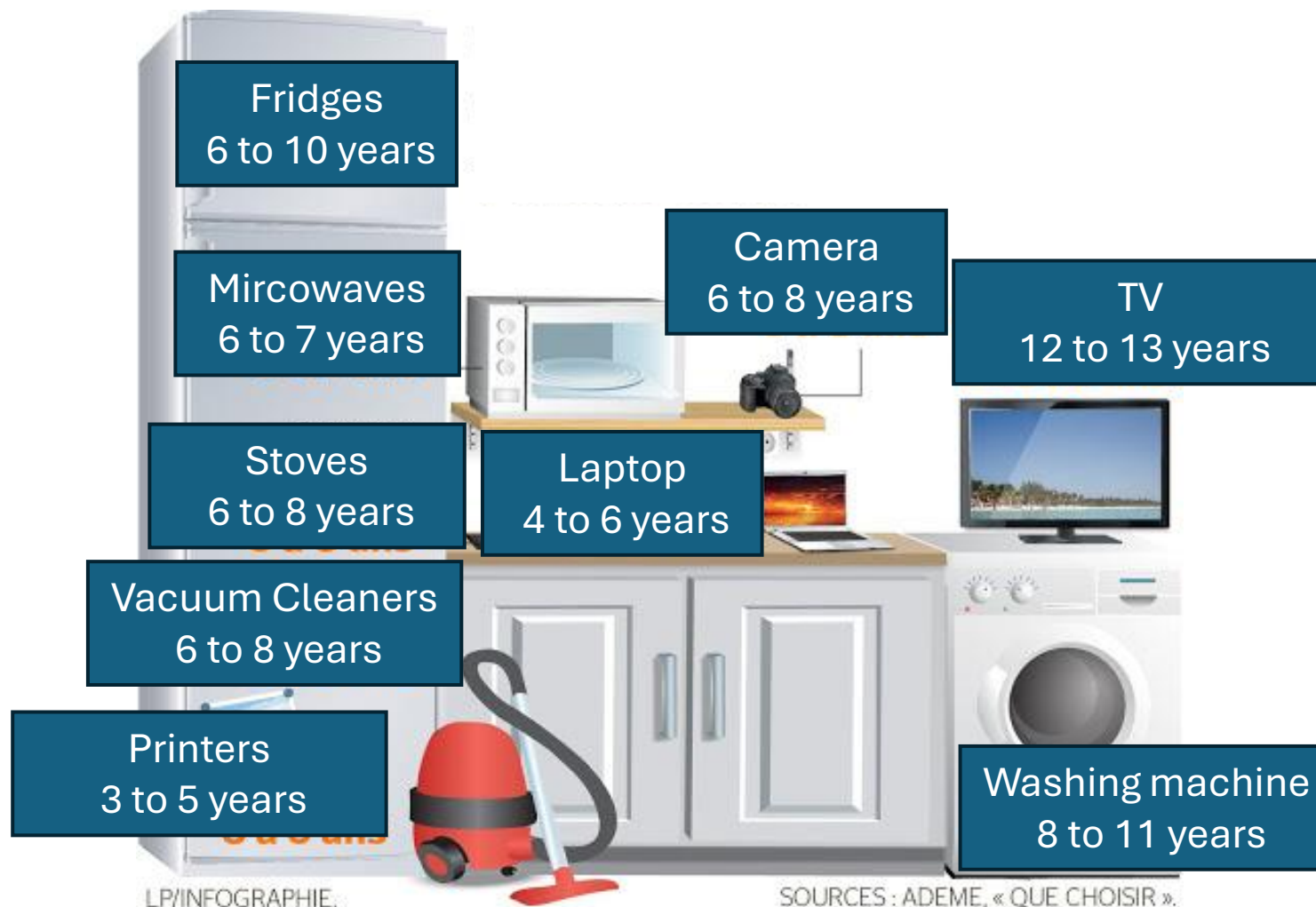
➔ Shelf life to be monitored

➔ Over-production to be avoided



Product Characteristics

Lifespan (reliability) -> Life cycle management



Product Characteristics

Production volume: large, medium or small series, or even unit production



Cars :
50 per hour



Plane :
50 per month



Cans:
50 per minute



Pilules :
50 per second

Truck :
50 per day



With multiple SKUs how to focus

ABC analysis

Most companies have a wide range of products. Should they be treated the same ? NO

ABC analysis is an arbitrary method to partition the products

- A category :
 - the happy few which generates 80% of the revenue;
 - they will need a lot of attention given the impact they have
- B category:
 - They generate the next 10% of the revenue
 - They are appropriate for some level of automation in decision making
- C category and beyond:
 - The rest
 - To be handled with the least attention

With multiple SKUs how to focus

ABC analysis

Total revenue: 154000 USD

Product	Quantity	Revenue
Tata1	2000	1000
Tata2	1000	2400
Tata3	3000	1300
Tata4	500	4500
Tata5	1500	2200
Tata6	1800	1100
Tata7	700	1800
Tata8	1000	500
Tata9	500	600



Product	Quantity	Revenue
Tata4	500	4500
Tata2	1000	2400
Tata5	1500	2200
Tata7	700	1800
Tata3	3000	1300
Tata6	1800	1100
Tata1	2000	1000
Tata9	500	600
Tata8	1000	500



The 1st 5 covers 80% of revenue

Is it sufficient to only look at the revenue contribution to classify the products?

Segmentation: Pull vs Push

- Push –
 - Execution is performed before the order is received
 - Demand is forecasted
 - Process offers shorter lead time to meet customer needs
- Pull –
 - Execution is performed when the order is in
 - Demand is actual and known with certainty
 - Reactive process based on actual need/demand

Manufacturing strategy:

Segmentation: MTS/MTO/ATO/ETO



	Design	Buy	Make	Assemble	Deliver
ETO	Design	Buy	Make	Assemble	Deliver
MTO		Inventory	Make	Assemble	Deliver
ATO		Inventory	Inventory	Assemble	Deliver
MTS		Inventory	Inventory	Inventory	Inventory

	Design	Buy	Make	Assemble	Deliver
ETO	Customer Lead Time				
MTO	Internal Lead Time		Customer Lead Time		
ATO	Internal Lead Time			Customer Lead Time	
MTS	Internal Lead Time				

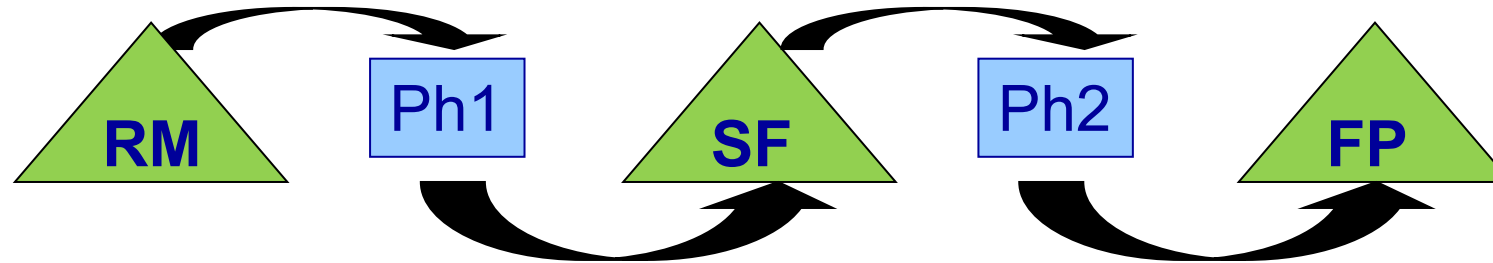
Segmentation: MTS/MTO/ATO/ETO



- Take a look at the website of your assigned company:
- Try to understand the supply chain model your company operates with
- Assign a manufacturing strategy to each (MTS/MTO/ATO/ETO)
- Present and explain your findings



Segmentation: MTS/MTO/ATO/ETO



Increase in cost by increase in added value



Raise the stock as far upstream as possible in the supply chain



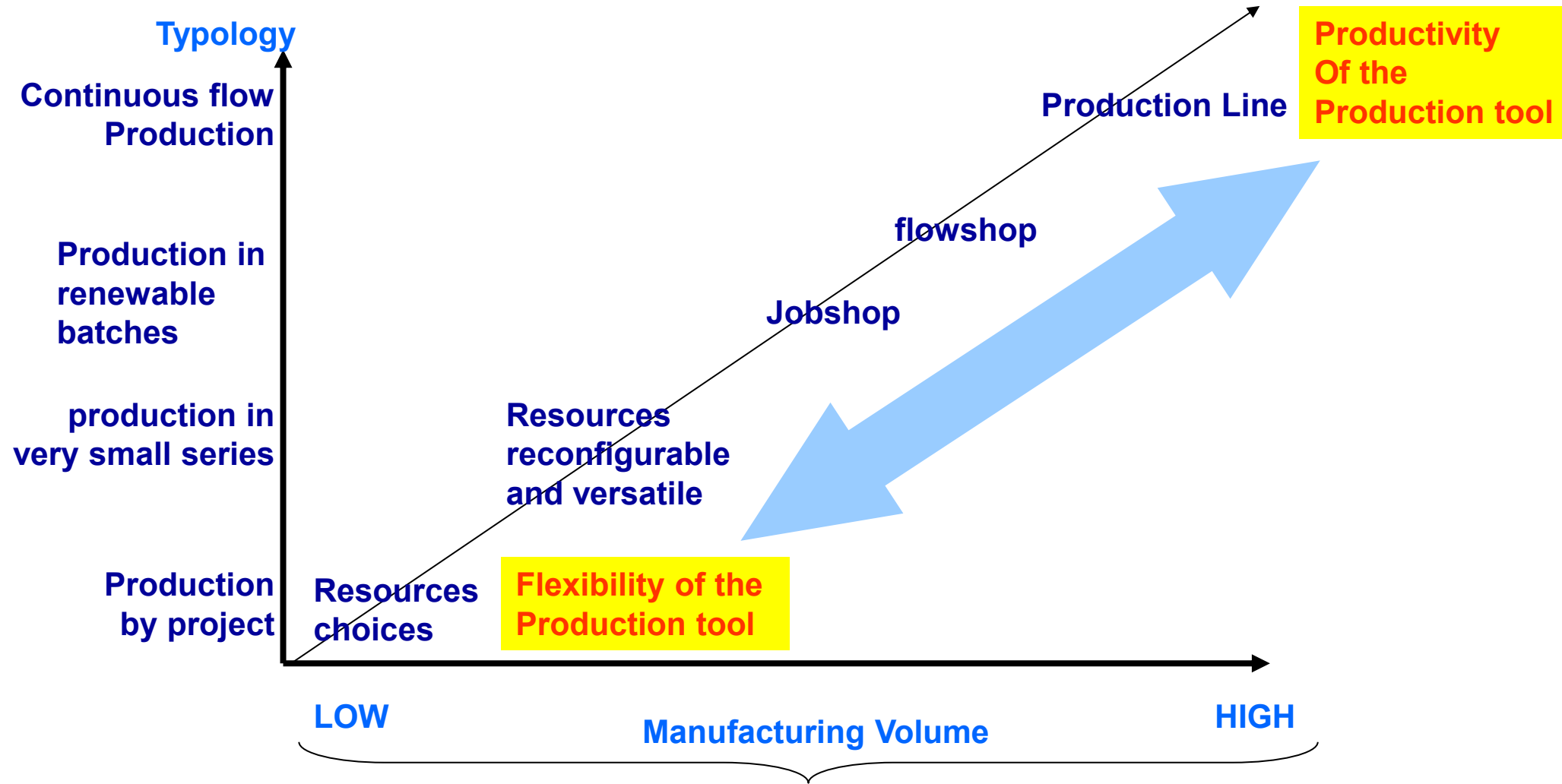
Decreased flexibility by item specialization



Reduce the variety of products in stock: late differentiation



Manufacturing strategy: Line Concept



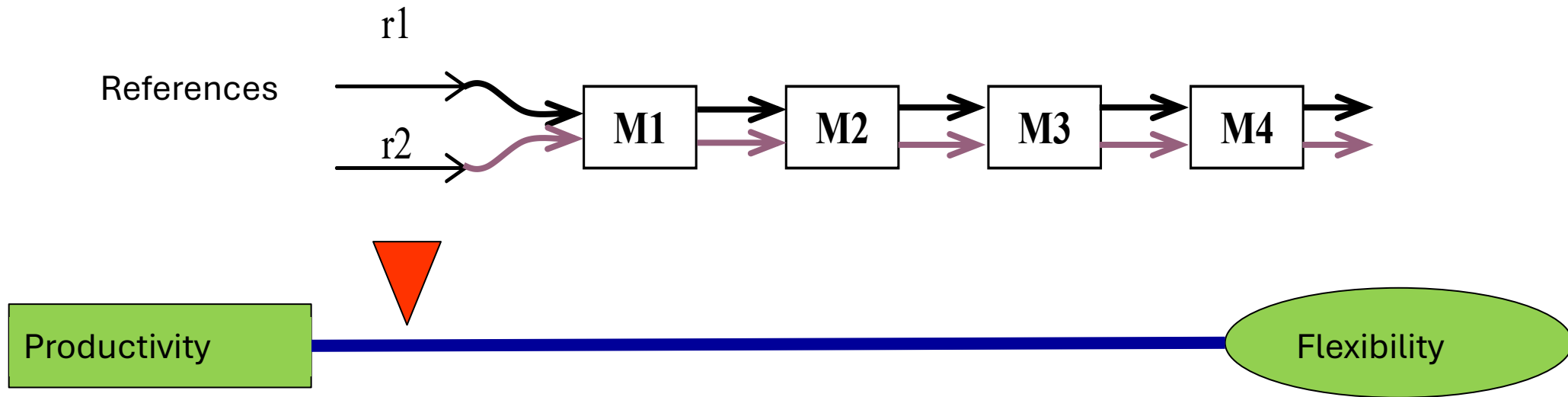
Production systems: high volume low mix

The production line

ex Manufacture of aluminum cans

Workstations set up according to the common sense of production In line, in U, in L...

The products all pass through each of the workstations

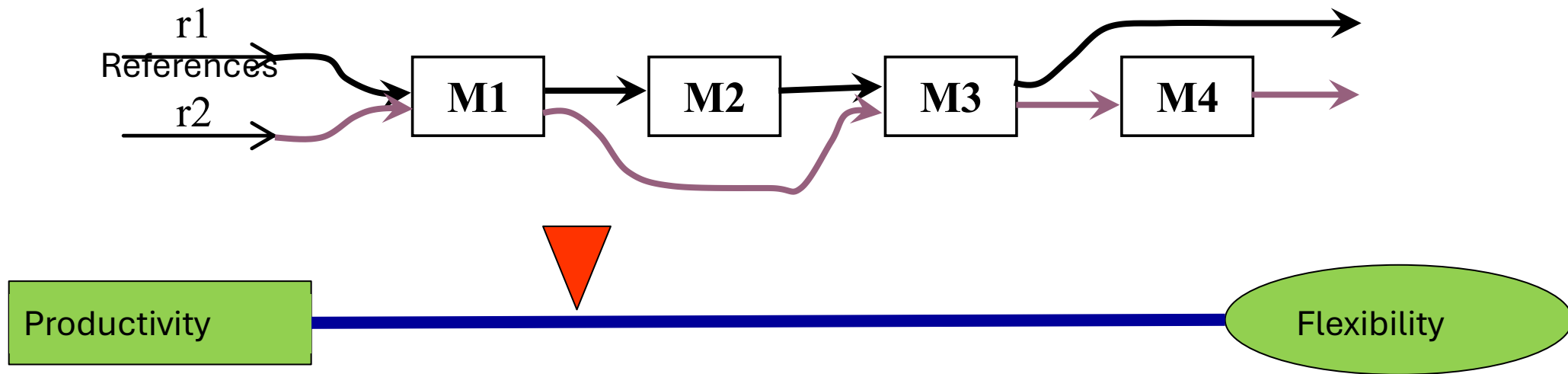


Production systems: high mix high volume

The flow shop:

Workstations set up according to the common sense of production In line, in U, in L...

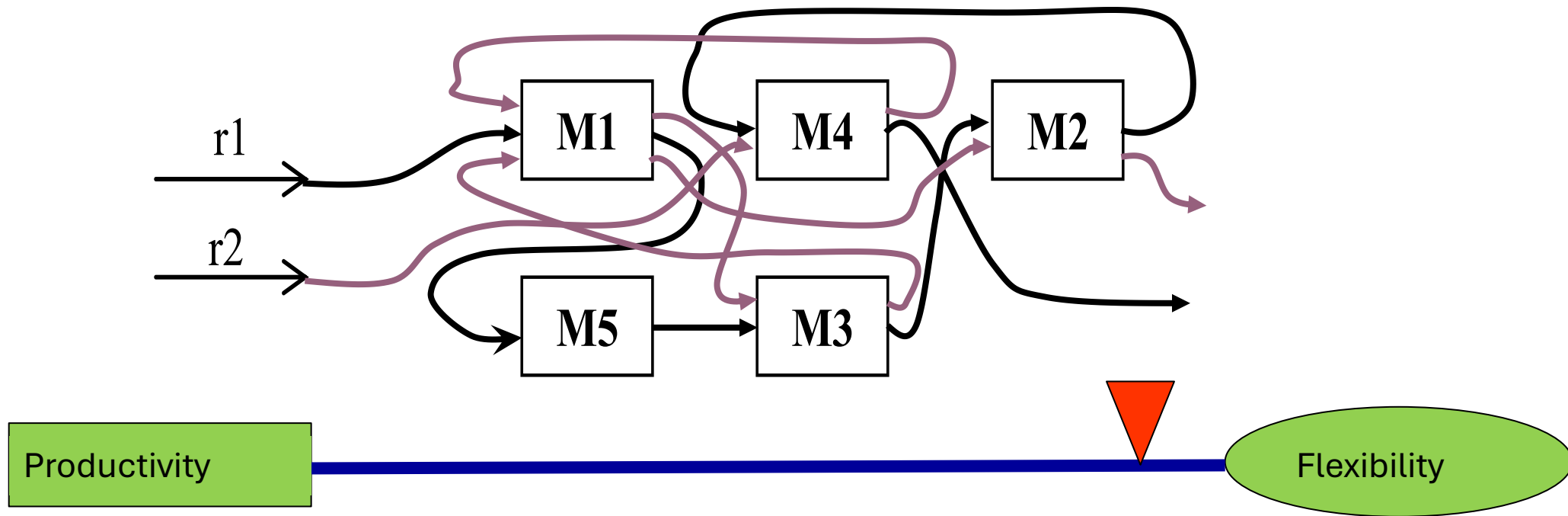
The products do not all pass through each of the workstations or machine adjustments are needed



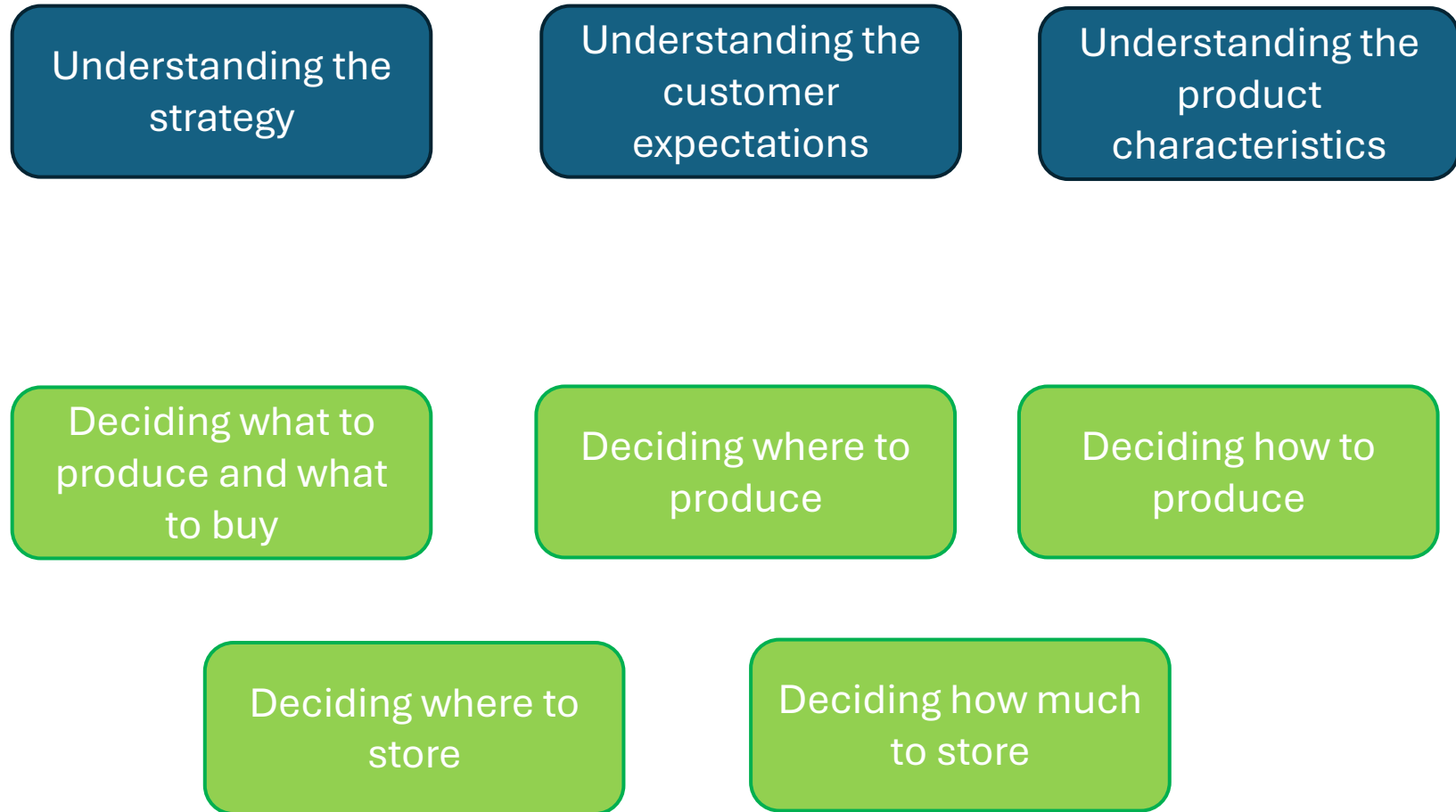
Production systems: high mix low volume

The job-shop:

Workstations grouped by technology or implemented “at best”



Summary



Capacity: how to define it?



The number of products produced per hour



The maximum number of products that can be put in a stock



The number of seats in a room (cinema, show, stadium, course, etc.).

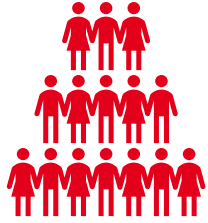


The number of hours available in a day



The number of people working on a project, etc

Capacity Planning



Short term

Human resources :

- Overtime
- Number of teams
- Interim

External resources:

- Subcontracting

Process improvement



Mid Term

Human resources :

- hiring
- Training / versatility

Technical resources:

- Investments in new machine
- Process improvements

External resources:

- partnership



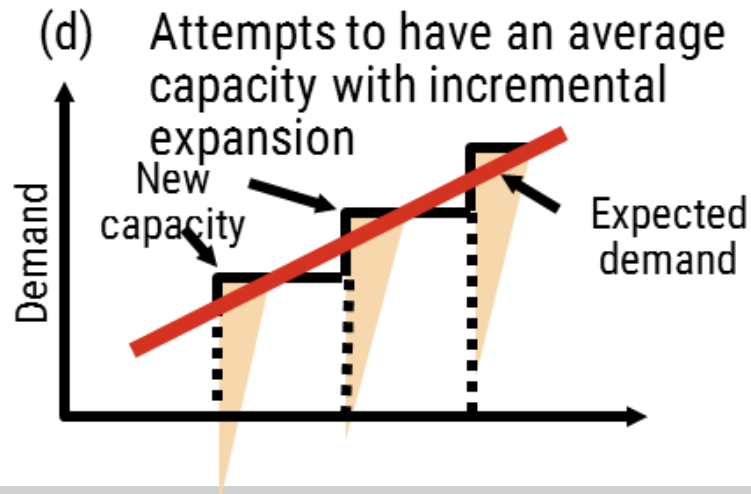
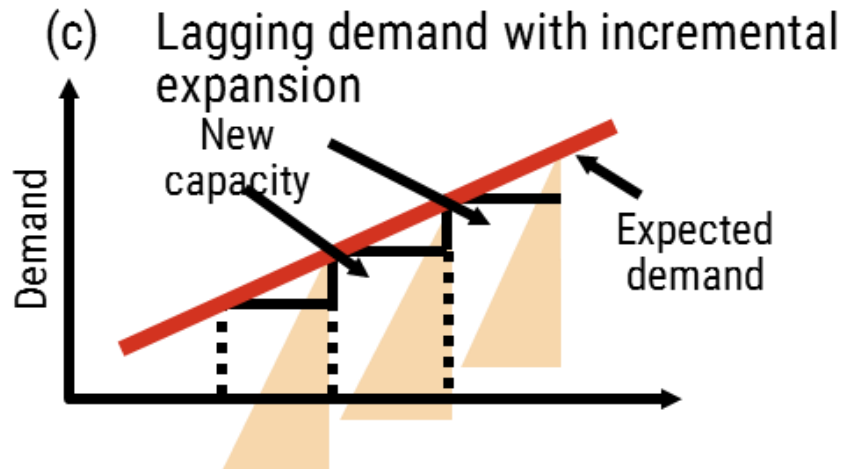
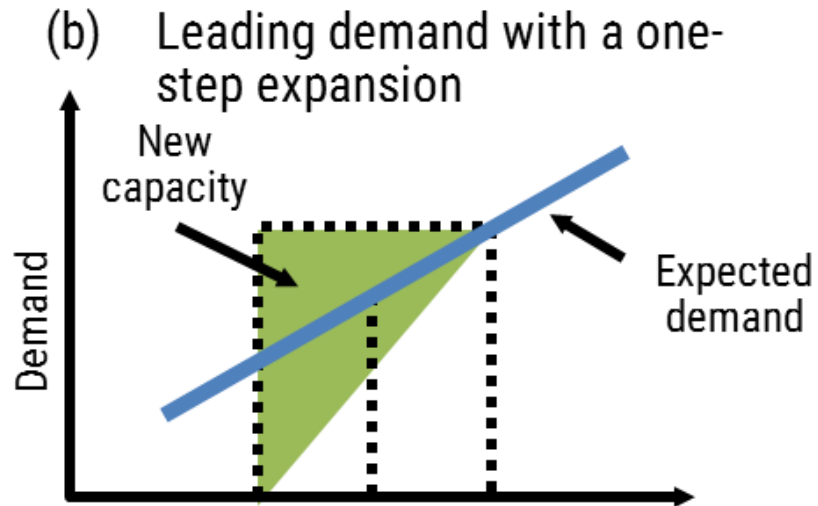
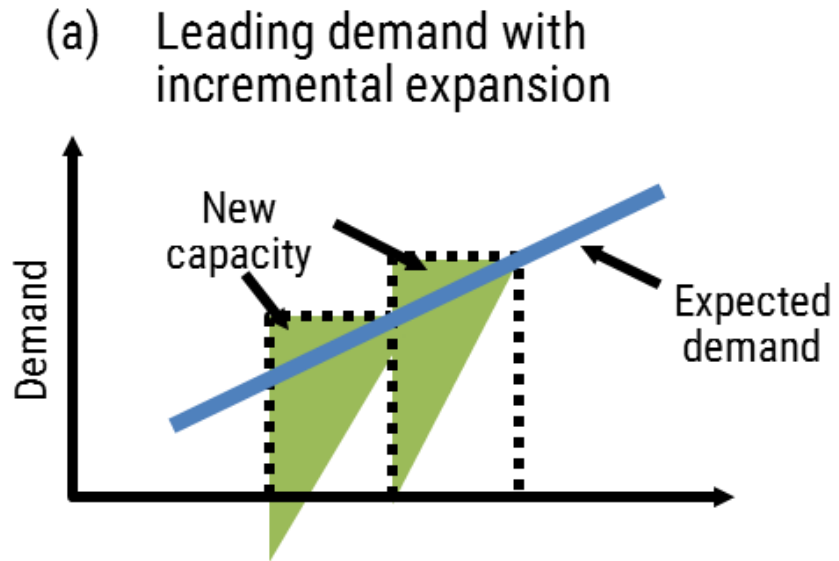
Long term

New Site

Site extension

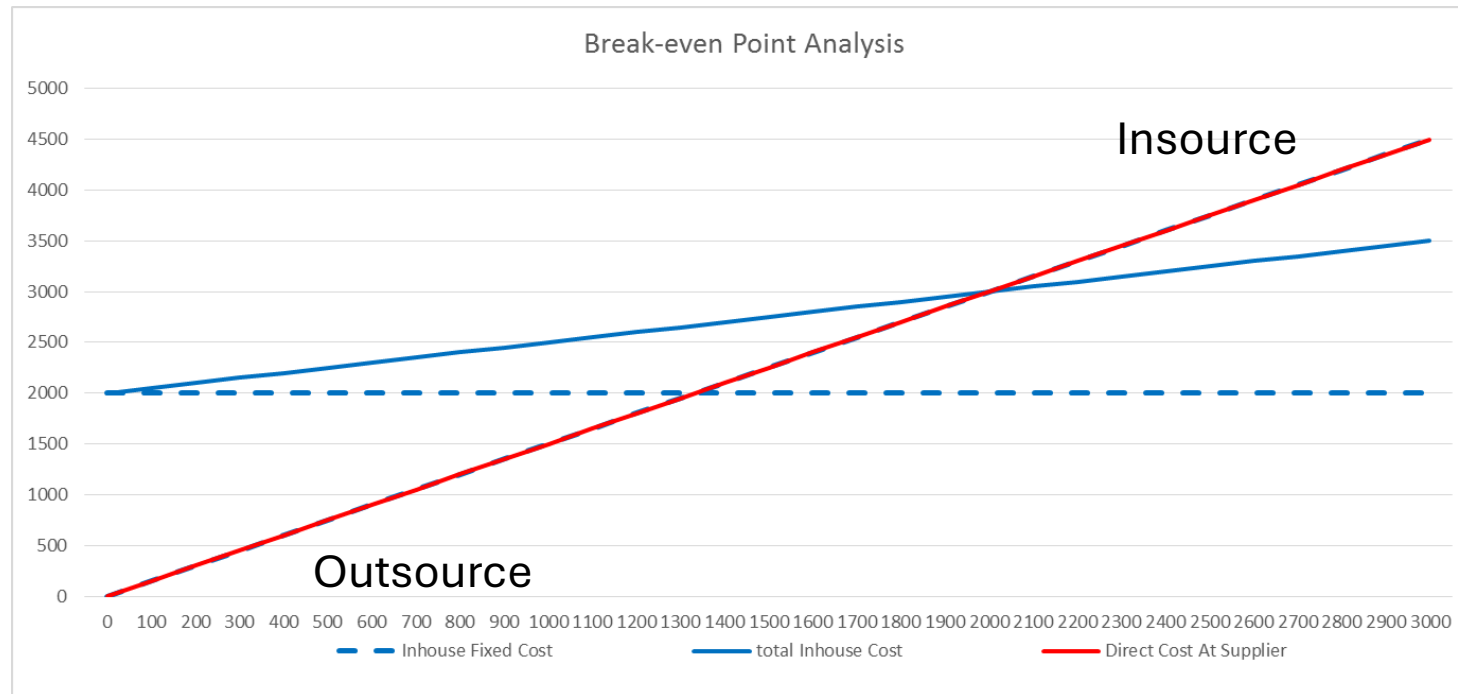
Investments

Capacity Planning: different strategies

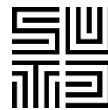


Capacity Planning: break-even analysis

- Typical example for make or buy:



Buy from some outside source: c_1/unit
 Internal production: c_2/unit
 Set-up or Investment cost $\$K$ to expand
 Cost for c_1/unit products BUY c_1x
 MAKE $K + c_2x$
 Breakeven point:
 $x = K/(c_1 - c_2)$



Readings and exercise

We have a guest speaker next week: do not be late for class!!

Do have a look at :

[o9 Solutions - Integrated Planning Powered by AI](#)

THANK YOU

TRAILBLAZING A BETTER WORLD BY DESIGN.

